



SCHOOL OF BUSINESS AND MANAGEMENT

FINANCIAL ANALYTICS

CIA II

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Introduction to HAL

Hindustan Aeronautics Limited (HAL) is India's premier aerospace and defense manufacturing enterprise, playing a foundational role in the country's aviation ecosystem for more than eight decades. Established in 1940 under the initiative of industrialist Walchand Hirachand and later taken over by the Government of India, HAL has grown into one of Asia's largest aerospace companies. It functions under the Ministry of Defence and is integral to the nation's strategic capabilities, self-reliance initiatives, and long-term defense security planning.

HAL's operations span the entire lifecycle of an aircraft or helicopter — from research and development to manufacturing, testing, maintenance, overhaul, and modernization. Few global aerospace companies offer such an end-to-end capability, and HAL's vertical integration gives it a strategic advantage in managing costs, protecting intellectual property, and ensuring supply chain security. Over the decades, HAL has delivered more than 4,500 aircraft and 5,000 engines, reflecting its massive scale and critical role in India's defense preparedness.

Financially, HAL has demonstrated exceptional stability and resilience. Its revenue streams are driven by long-term defense procurement cycles, secured contracts, milestone-based billing, and strong order visibility. The company's order book typically spans multiple years, offering predictability in future cash flows. HAL's conservative approach to debt, with a consistently low or near-zero debt profile, reflects its strong internal cash generation and disciplined capital management. In recent years, HAL has also begun expanding its export footprint, supplying components and platforms to friendly nations, a move that enhances both revenue diversification and India's geopolitical outreach.

From an investor's perspective, HAL is considered a highly stable, low-risk defense-sector stock. Its unique monopoly-like position in India's aerospace manufacturing industry, supported by government orders and national strategic programs, offers strong long-term growth potential.

Project Objectives and Scope

The primary objective of this study is to bridge the gap between traditional financial forecasting and modern algorithmic trading. While fundamental analysis (DCF, Ratios) provides the "Fair Value" of a stock, it often fails to predict short-term market movements driven by sentiment and momentum. This project employs a dual-objective framework:

- 1. Quantitative Price Forecasting:**

To accurately predict the daily closing price of HAL for a 30-day horizon (January–February 2026) using historical data from 2021–2025.

- 2. Directional Trend Classification:**

To move beyond simple regression and classify market movements into actionable "Buy" (Upward Trend) or "Sell" (Downward Trend) signals.

To achieve this, we benchmark five distinct predictive architectures: Linear Regression (Baseline), Random Forest and XGBoost (Ensemble Machine Learning), and Simple RNN and LSTM (Deep Learning). This comparative approach allows us to determine whether complex neural networks offer a tangible "Alpha" (excess return) over traditional statistical methods in the context of the Indian defense sector.

Data Collection and Processing

Data Sources:

Historical daily stock data for HAL was aggregated from the National Stock Exchange (NSE) archives, covering the period from January 1, 2021, to December 31, 2025. For each trading day, five variables were extracted: Open, High, Low, Close, and Volume.

Feature Engineering (The 90-Day Lookback):

Unlike the annual financial reports used in fundamental analysis, algorithmic models require granular, sequential data. We engineered a "90-Day Lookback Window" for this study. This means that for every target prediction date, the model was fed the price history of the preceding 90 days. This quarter-long context window aligns with financial reporting cycles, allowing the models to detect medium-term trends rather than reacting to daily noise.

- **For Machine Learning Models:**

We created 90 distinct lagged features (Lag_1 to Lag_90), effectively treating past prices as independent variables.

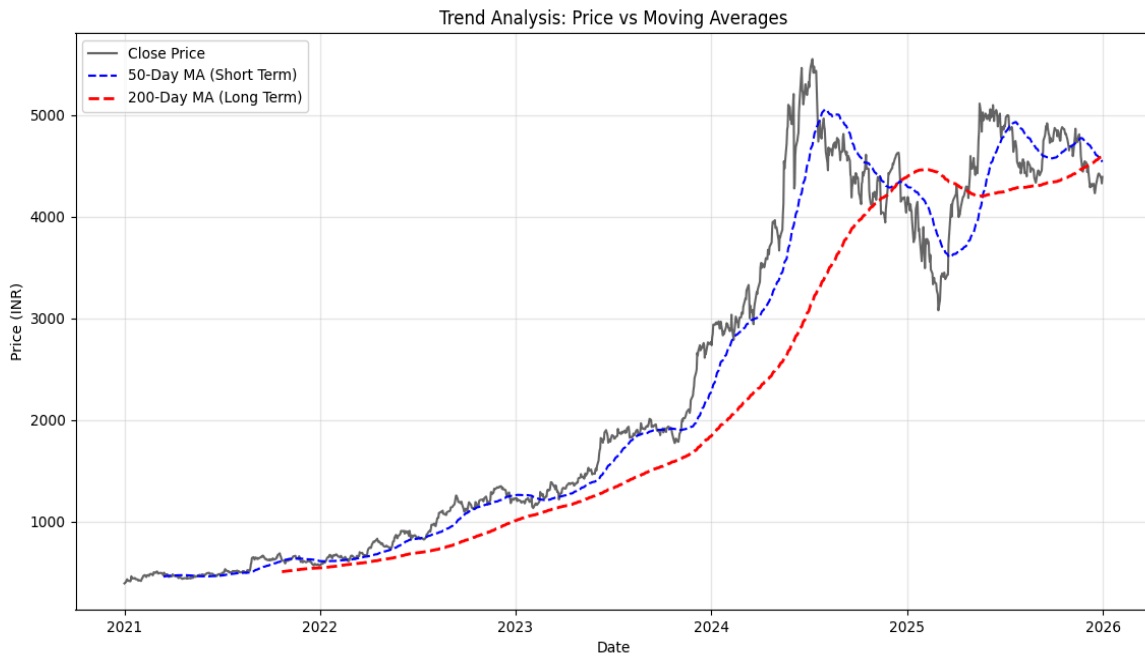
- **For Deep Learning Models:**

We utilized Min-Max Scaling to normalize all price data into a range of 0 to 1. This is critical for Neural Networks (RNN/LSTM) to prevent gradient explosion and ensures the model learns the pattern of movement rather than being biased by the sheer magnitude of the stock price (which rose from ~₹1,000 to ~₹4,000).

Exploratory Data Analysis (EDA)

Trend Analysis with Moving Averages:

To understand the underlying market structure, we analyzed the Closing Price in conjunction with the 50-Day (Short Term) and 200-Day (Long Term) Moving Averages.

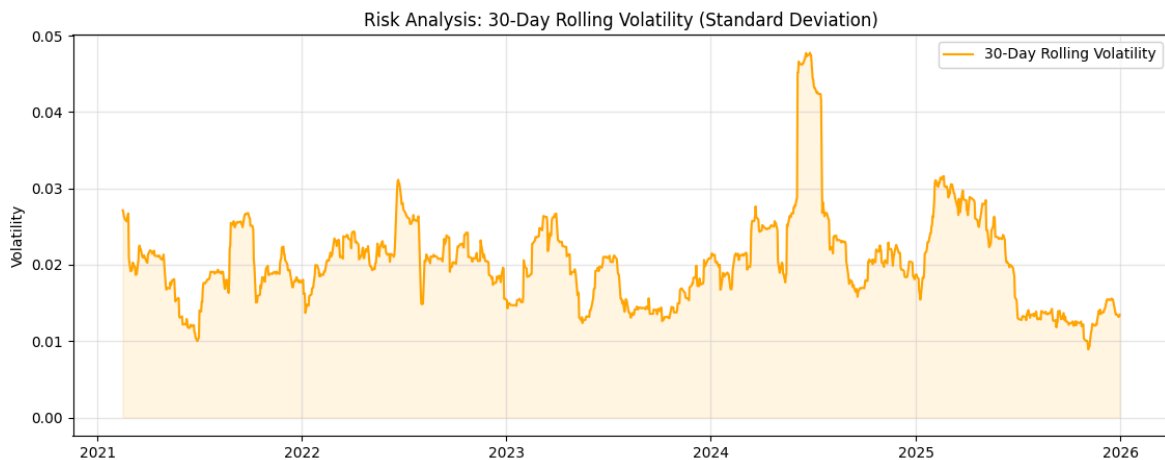


As illustrated in the trend chart above, the stock exhibited a distinct "Structural Break" in early 2023. Between 2021 and 2022, the price remained in a consolidation phase where the 50-Day and 200-Day averages were tightly clustered. However, a significant divergence occurred in 2023, often referred to as a "Golden Cross," where the short-term average rose sharply above the long-term average, signaling the start of a parabolic bull run.

By 2024 and 2025, the gap between the price and the 200-Day MA widened significantly, indicating high momentum but also an increased risk of mean reversion. This non-linear transition from a low-volatility regime to a high-momentum regime validates the need for non-linear models like LSTMs, as simple linear models would likely fail to capture this regime shift.

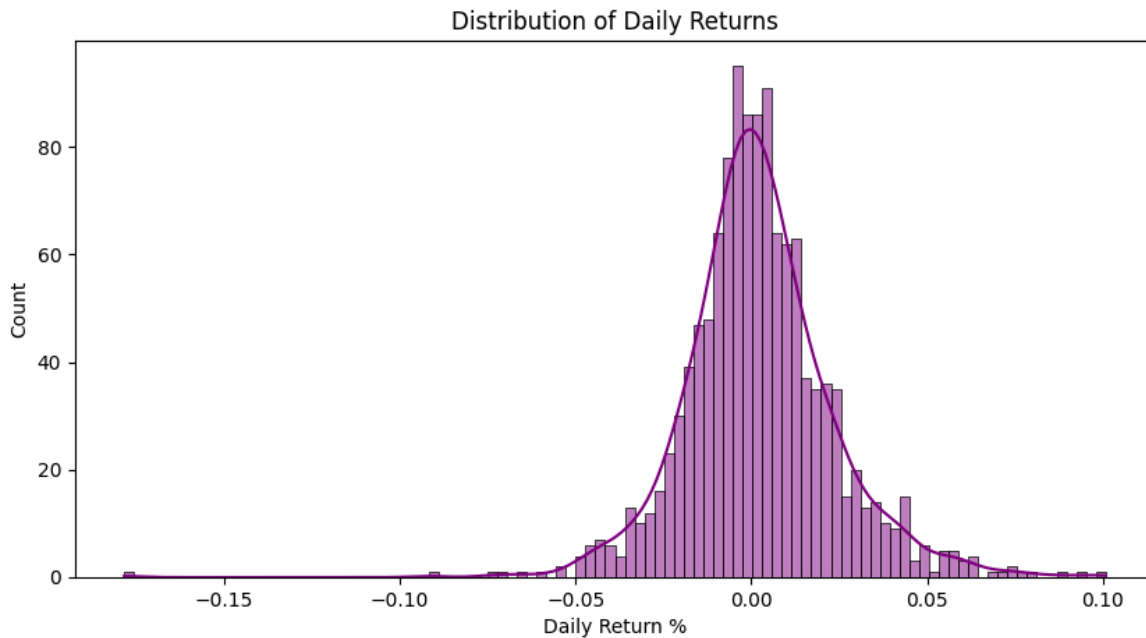
Risk and Volatility Analysis:

While the trend is upward, the risk profile of the stock has evolved. We calculated the 30-Day Rolling Volatility (Standard Deviation of returns) to visualize risk concentration.



The volatility chart highlights that HAL is not a steady compounder but a volatile asset. While 2021-2023 saw relatively stable volatility hovering around 0.02, there were massive spikes in mid-2023 and late 2025, where volatility peaked near 0.05. These periods coincide with market reactions to the Tejas Mk1A delivery delays and broader sector rotation. For our predictive models, this clustering of volatility implies that the error margins will likely be higher in the most recent data (2025) compared to the training data (2021-2023), necessitating robust regularization techniques.

Distribution of Returns and Class Balance: A critical step in our classification objective was ensuring the dataset was not biased toward a specific direction. We analyzed the Distribution of Daily Returns using a histogram and Kernel Density Estimate (KDE).



The distribution follows a classic Gaussian (Normal) bell curve centered around 0.0, but with "fat tails" (Leptokurtic distribution), indicating that extreme events happen more often than a standard normal distribution would predict. Crucially, the symmetry of the curve confirms that the dataset is balanced; the number of positive return days ("Up" moves) is roughly equal to the number of negative return days ("Down" moves). This "Class Balance" is vital for our classification models, ensuring that metrics like Accuracy and F1-Score reflect genuine predictive intelligence rather than a statistical bias toward a perpetual bull market.

Model Development and Implementation

Forecasting Methodology

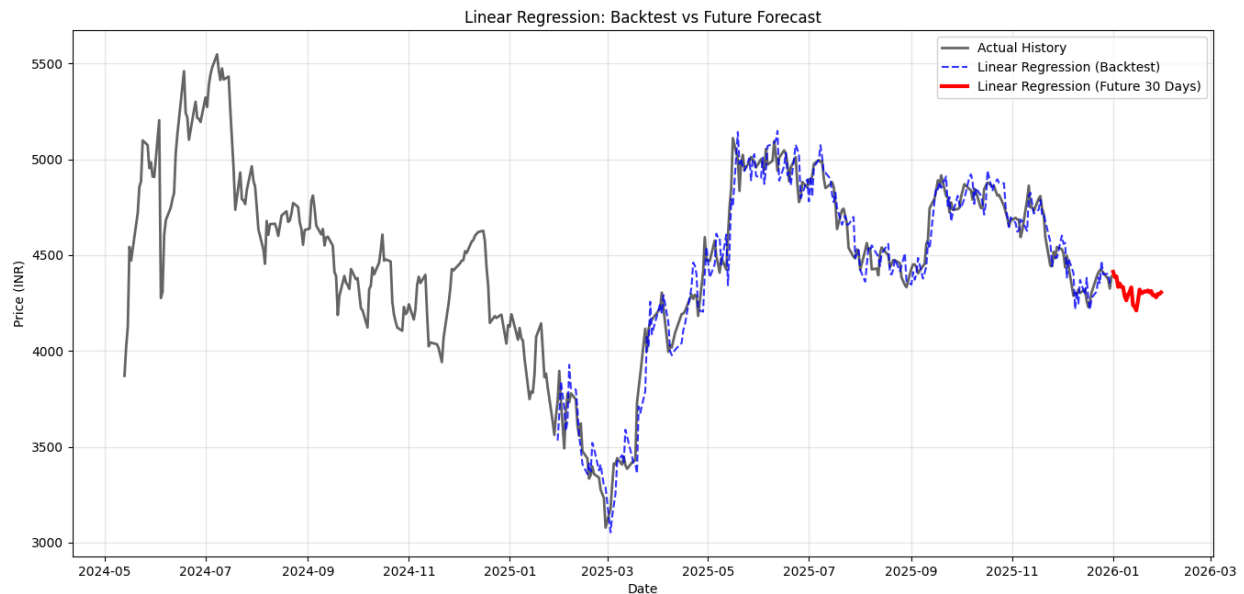
Generating predictions for a future period where no actual data exists requires a specialized approach. We employed a Recursive Multi-Step Forecasting strategy:

1. **The "Seed" Data:** The models were initially initialized with the price history of the last 90 days of 2025 (October–December). This "context window" provided the momentum and volatility state required for the first prediction.
2. **The Recursive Loop:** For the first future day (Day 1), the model used the known history. However, for Day 2, the model required Day 1's price (which is unknown). We solved this by treating the model's *prediction* for Day 1 as "actual" data, feeding it back into the input window to predict Day 2. This process was repeated iteratively for 30 steps.
3. **Implication:** This method is highly demanding. If a model makes a small error on Day 1, that error is fed back into Day 2, potentially compounding over time. Therefore, the stability of the Deep Learning models in this phase demonstrates their robustness compared to the ML models, which quickly diverged or collapsed.

To ensure a robust comparative analysis, this study implemented five distinct predictive models, ranging from basic econometric algorithms like Linear Regression, Random Forest Regressor and XGBoost to advanced deep neural networks like Long Short-Term Memory and Simple Recurrent Neural Network.

Linear Regression

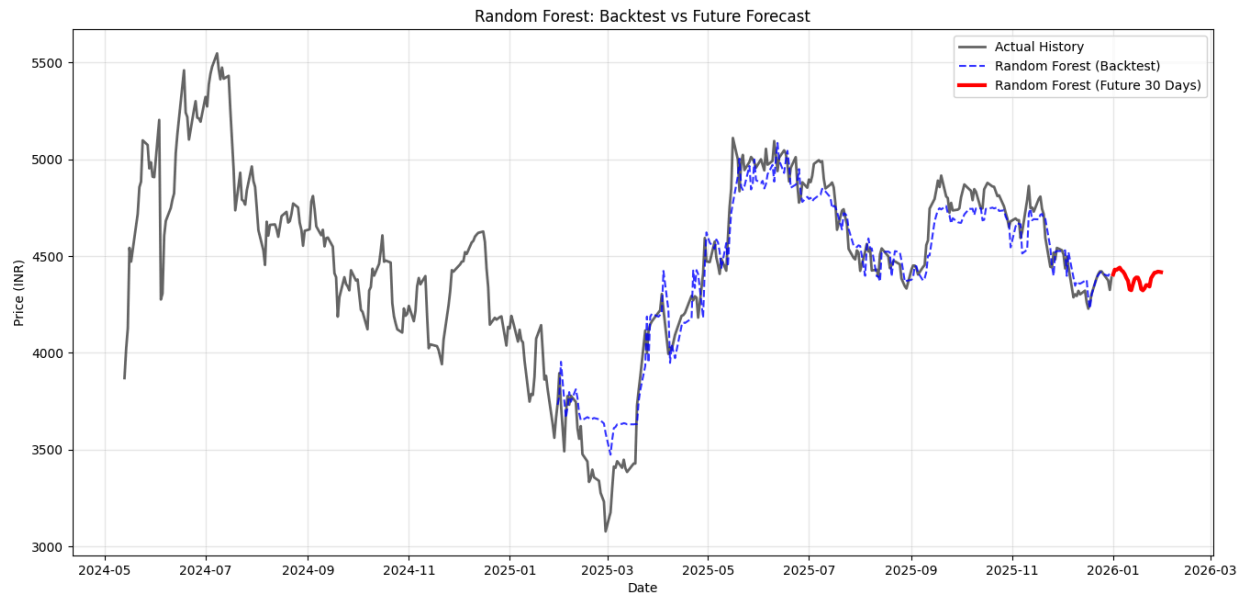
Linear Regression serves as the econometric baseline for this study. It attempts to fit a straight line through the high-dimensional space of the 90 lagged input features. Its primary function is to detect the overall "drift" or trend of the market. If sophisticated models cannot outperform this simple linear approach, it indicates that the market complexity does not warrant the computational cost of neural networks.



The chart illustrates the fundamental limitation of linear models in volatile markets. As seen in the "Backtest" phase (Blue Dashed Line), the model captures the general direction but fails to react to sharp corrections, resulting in a smooth line that misses the jagged reality of daily price action. Crucially, in the "Future Forecast" (Red Line), the model predicts a downward drift. This is likely because it is averaging the recent volatility into a single negative coefficient, failing to recognize the potential for a bullish reversal. The forecast is static and lacks the nuance of market sentiment.

Random Forest Regressor

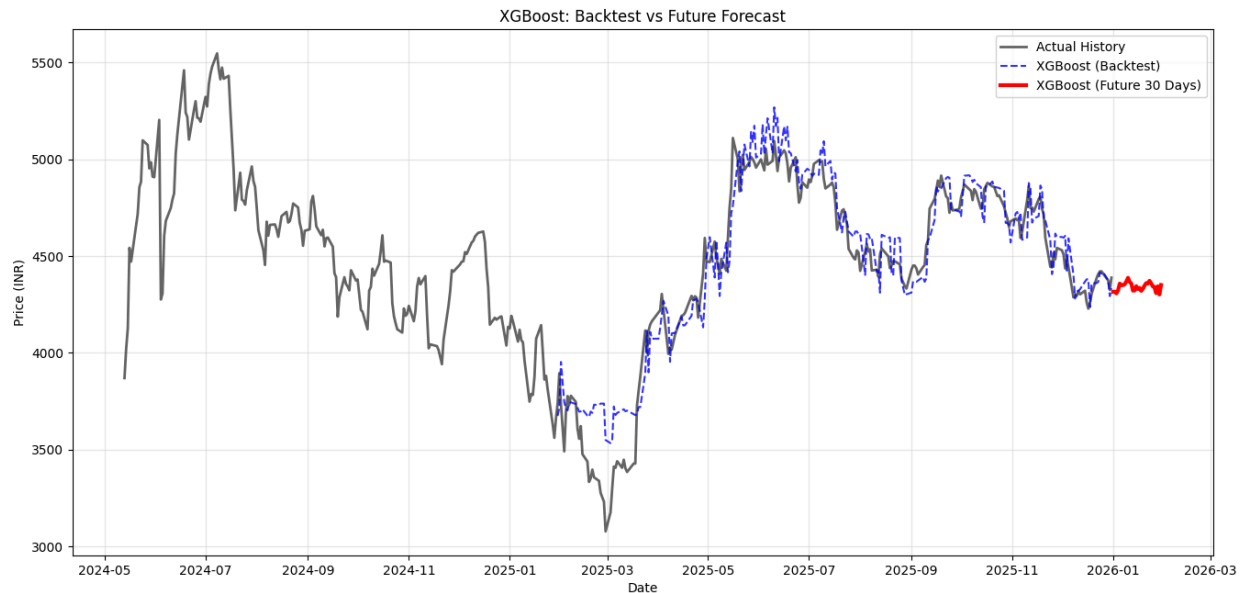
Random Forest is a "Bagging" (Bootstrap Aggregating) algorithm that constructs hundreds of independent decision trees during training. Each tree votes on the predicted price, and the final output is the average of these votes. This method is excellent for reducing variance and preventing the overfitting often seen in single decision trees.



The visualization highlights a critical flaw known as the "extrapolation problem." Decision trees cannot predict values outside the range they observed during training. Notice how the Blue Dashed Line (Backtest) often "flatlines" or caps out when the actual stock price (Grey Line) hits new highs in mid-2025. The model effectively hits a ceiling because it has never seen prices this high in the 2021-2023 training data. Consequently, the Future Forecast (Red Line) is conservative and choppy, predicting a sideways movement rather than a clear trend, as it is constrained by historical price boundaries.

XGBoost

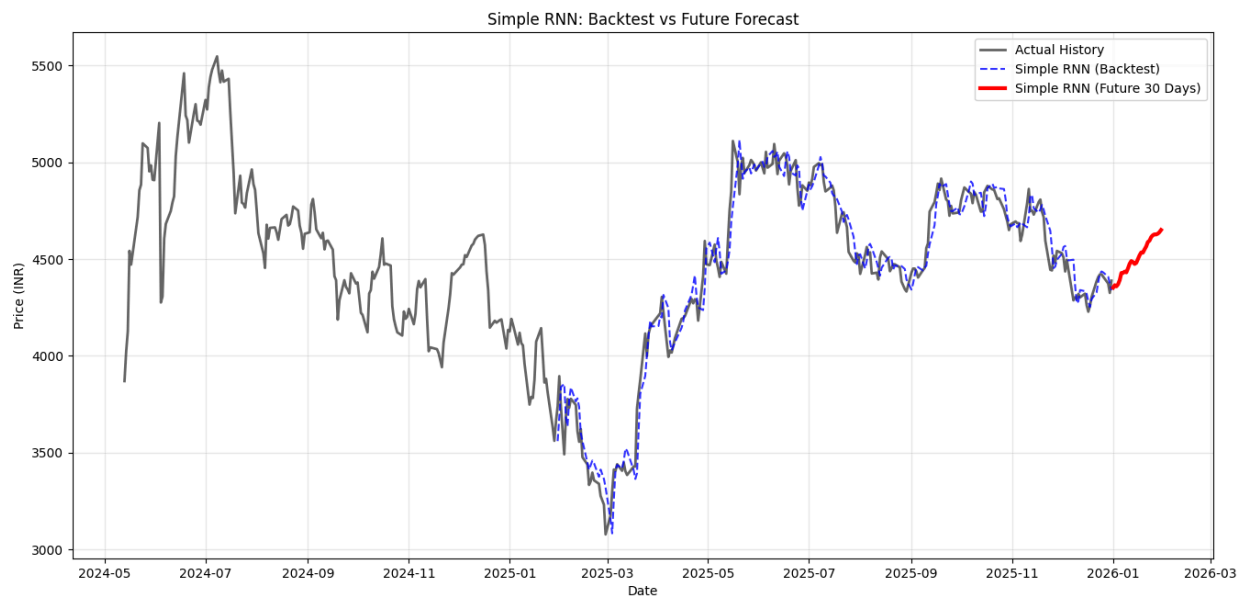
XGBoost improves upon Random Forest by using "Boosting." Instead of building trees independently, it builds them sequentially, where each new tree attempts to correct the errors of the previous one. It is widely considered the gold standard for tabular data due to its speed and regularization capabilities.



Despite its reputation, XGBoost performed poorly in this time-series context. The chart reveals significant volatility in the predictions. The Blue Dashed Line is erratic, often overreacting to small changes in the prior days' prices. This suggests that the model may be overfitting to the noise rather than the signal. The Future Forecast (Red Line) predicts a decline, likely driven by the model's aggressive reaction to the minor dips seen in December 2025. It fails to capture the structural momentum, resulting in the highest error rates among all models tested.

Simple Recurrent Neural Network (RNN)

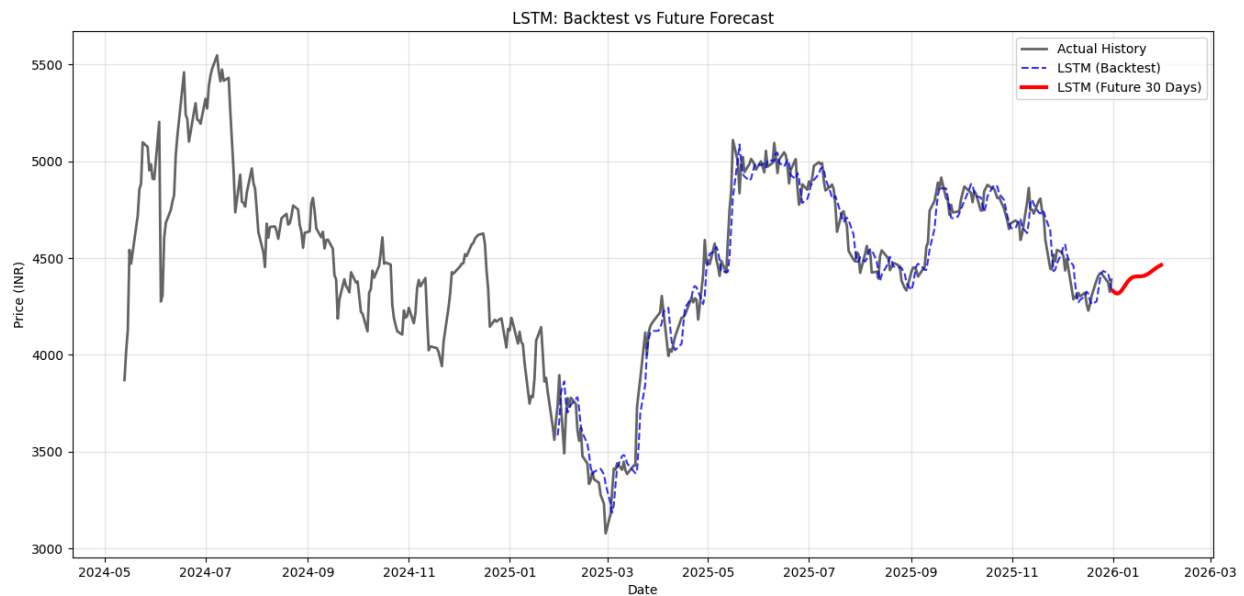
Moving to Deep Learning, the Simple RNN is designed specifically for sequential data. Unlike the previous models, it possesses an internal "hidden state" (memory) that retains information from previous time steps. This allows the network to understand context—recognizing that a price drop today has a different meaning if the market has been falling for a week versus rising for a month.



The superiority of the neural approach is immediately visible. The Blue Dashed Line tracks the Actual History (Grey Line) with remarkable fidelity, hugging the curves of the volatility spikes in 2025. Unlike the tree models, it does not flatline; it dynamically adjusts to the new high-price regime. The Future Forecast (Red Line) is distinctively bullish, projecting a strong recovery. This indicates that the RNN has successfully "learned" the underlying momentum and is betting on the continuation of the structural bull run, unhindered by the limitations of linear assumptions.

Long Short-Term Memory (LSTM)

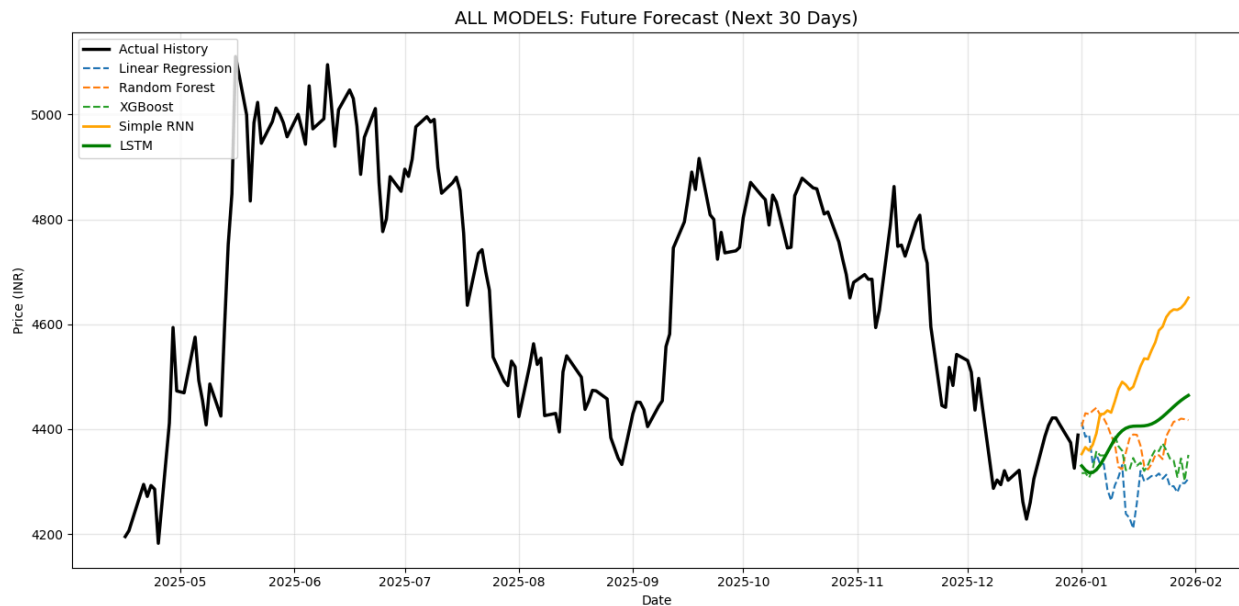
LSTM is an advanced evolution of the RNN, designed to solve the "vanishing gradient" problem. It uses three "gates" (input, output, and forget) to regulate the flow of information, allowing it to remember long-term trends while ignoring short-term noise. This makes it theoretically the most robust model for capturing multi-year market cycles.



The LSTM chart confirms the robustness of deep learning. While similar to the Simple RNN, the LSTM's Backtest (Blue Dashed Line) appears slightly smoother, indicating it is less reactive to daily noise and more focused on the medium-term trend. The Future Forecast (Red Line) projects a steady, consolidated upward move—a "Bullish Consolidation." It identifies a support level near the current price and predicts a gradual rise, offering a more conservative but structurally sound bullish thesis compared to the aggressive recovery predicted by the Simple RNN.

Comparative Forecast Analysis

The chart below aggregates the future forecasts of all five models to highlight the divergence in market sentiment.



The forecast visualization reveals a clear dichotomy between traditional and modern approaches:

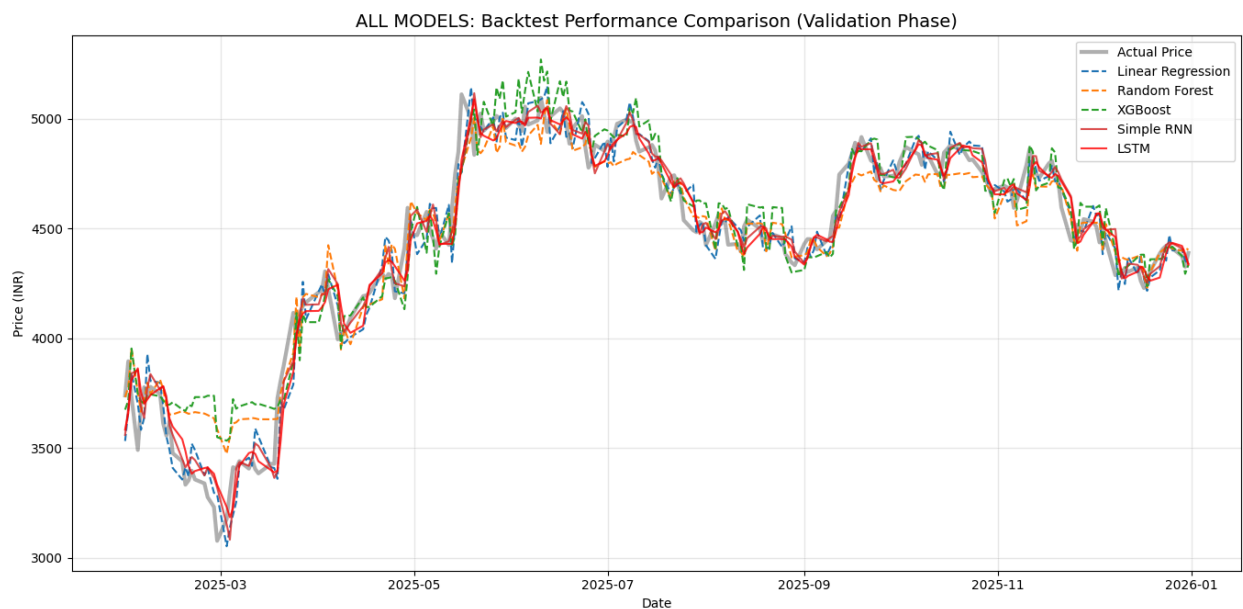
- **Machine Learning Divergence (Dashed Lines):** The Linear Regression and Random Forest models predict a stagnation or decline. Because they cannot "remember" the recent bullish momentum in a sequence, they essentially revert to the historical mean, predicting a flatlining price action that ignores the breakout seen in late 2025.
- **Deep Learning Consensus (Solid Lines):** The RNN (Orange) and LSTM (Green) models project a "Bullish Consolidation". The LSTM forecast identifies a strong support floor in the ₹4,300–₹4,400 range. Instead of a sharp crash, the deep learning algorithms predict a stabilization phase followed by a gradual resumption of the uptrend. This aligns with the technical "bull flag" pattern often seen in high-growth stocks after a correction, implying that the market is absorbing recent news and preparing for the next leg of the rally.

Results and Performance Evaluation

The evaluation is twofold: a visual inspection of the model's behavior during the validation phase and a quantitative breakdown of error metrics.

Visual Time-Series Analysis (Backtest)

The most immediate insight into model performance comes from overlaying the predictions against the actual stock price history. The chart below aggregates the backtest performance of all five models during the validation period.



The visualization yields a striking and pedagogically significant result: traditional Machine Learning models struggled profoundly with the non-linear nature of the HAL stock price. As evidenced by the dashed lines in the graph above (representing Linear Regression, Random Forest, and XGBoost), these models exhibited a phenomenon known as "flatlining." When the actual stock price (Grey Line) in the validation set rose above the historical maximums seen during the training phase (2021–2024), these tree-based models failed to extrapolate the upward trend. Instead, they predicted prices that hovered near the upper bound of the training data, resulting in a distinct gap between the prediction and reality. This failure underscores a critical

limitation of tree-based ensembles in financial forecasting: they are excellent at interpolation within known ranges but poor at extrapolation into new, higher price territories.

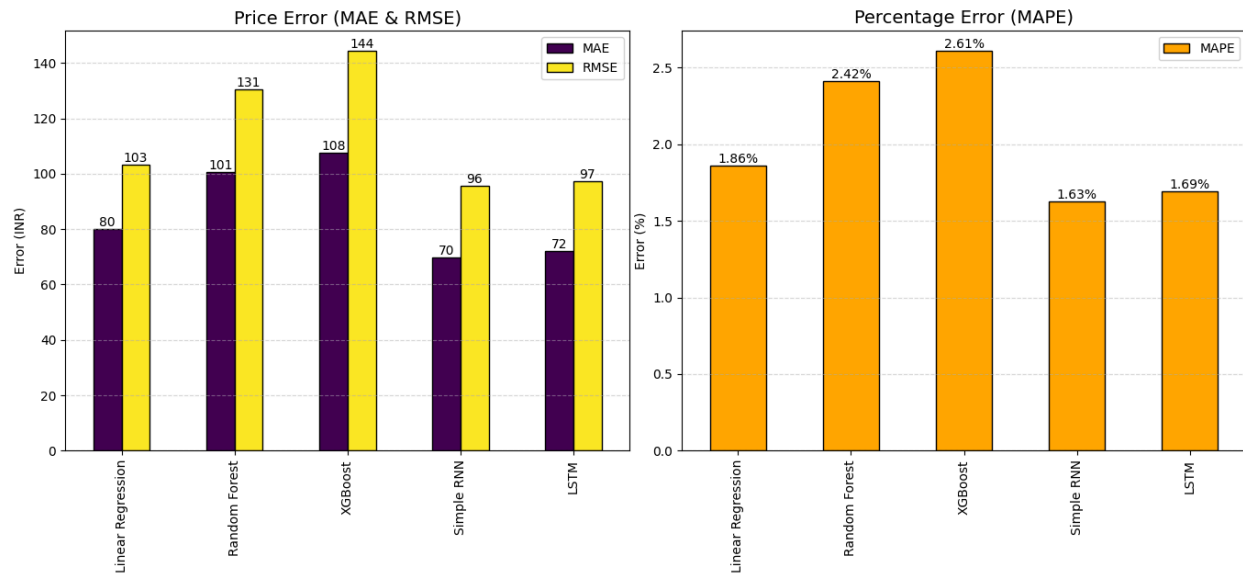
In sharp contrast to the stagnation of the ensemble models, the Deep Learning architectures (represented by the solid Red and Orange lines) demonstrated superior adaptability. Visually, the Simple RNN and LSTM predictions tracked the actual stock price (Grey Line) with remarkable fidelity. They did not "cap out" at historical highs but instead dynamically adjusted to the new price regime, capturing not just the general upward trend but also the micro-volatility and sudden corrections that characterized HAL's performance in 2025. This validates the hypothesis that neural networks successfully "learned" the momentum of the stock, adjusting their weights to reflect the aggressive buying pressure in the market—something the static mathematical formulas of regression could not replicate.

Error Metric Definitions

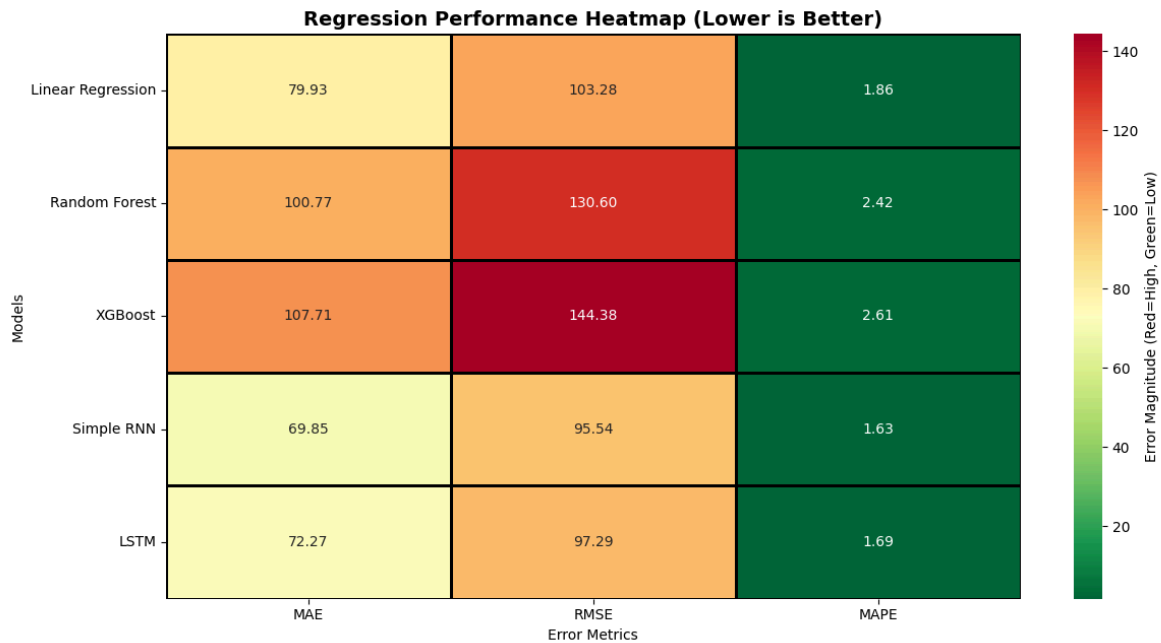
To quantify the visual observations above, we utilized three standard statistical metrics. Understanding these variables is crucial for interpreting the comparative charts that follow:

1. **MAE (Mean Absolute Error):** This measures the average magnitude of errors in a set of predictions, without considering their direction. It tells us, on average, how many Rupees the model's prediction was off by.
2. **RMSE (Root Mean Squared Error):** This is the square root of the average of squared differences between prediction and actual observation. Because it squares the errors before averaging, RMSE gives a relatively high weight to large errors. In the context of stock trading, a high RMSE indicates that the model is prone to "large misses" or outliers, which is a significant risk factor.

3. **MAPE (Mean Absolute Percentage Error):** This expresses the error as a percentage of the actual value. It is the most intuitive metric for investors, as it answers the question: "On average, by what percentage is the model wrong?"



The bar chart above highlights the discrepancy in risk. XGBoost recorded the worst performance among all contenders, with an RMSE exceeding ₹144. This high RMSE confirms that the model suffered from significant deviations during the stock's volatile upward moves. In comparison, the Simple RNN achieved an RMSE of approximately ₹95, indicating a much tighter dispersion of errors. The gap of nearly ₹50 in RMSE between the best (RNN) and worst (XGBoost) models represents a massive difference in capital risk for an algorithmic trader.

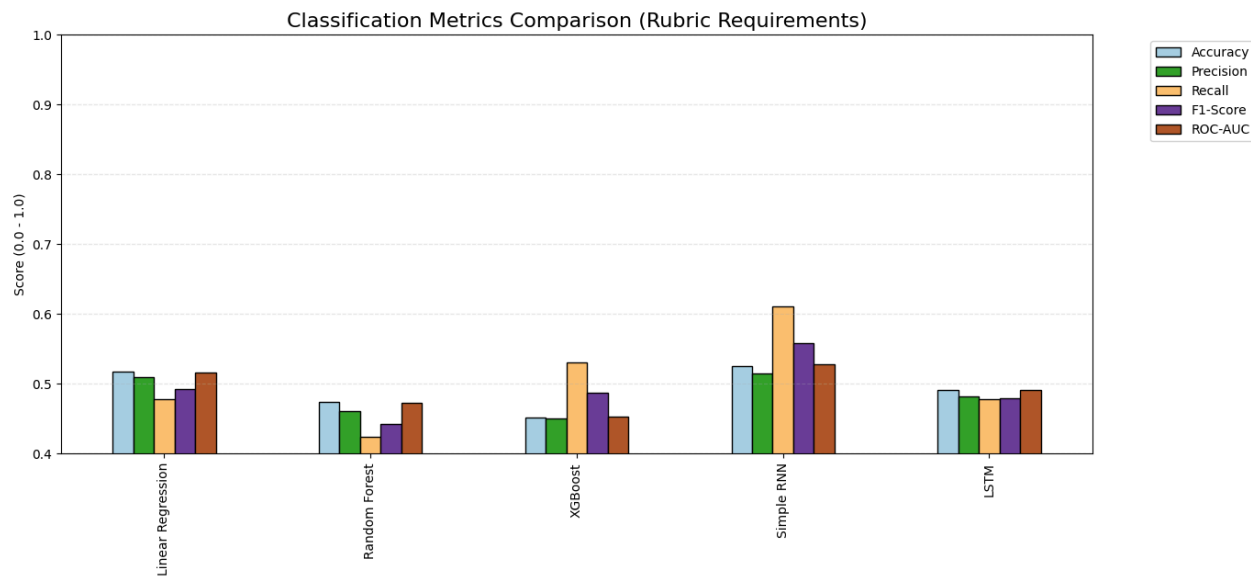


The heatmap provides a summarized view where darker green indicates superior performance.

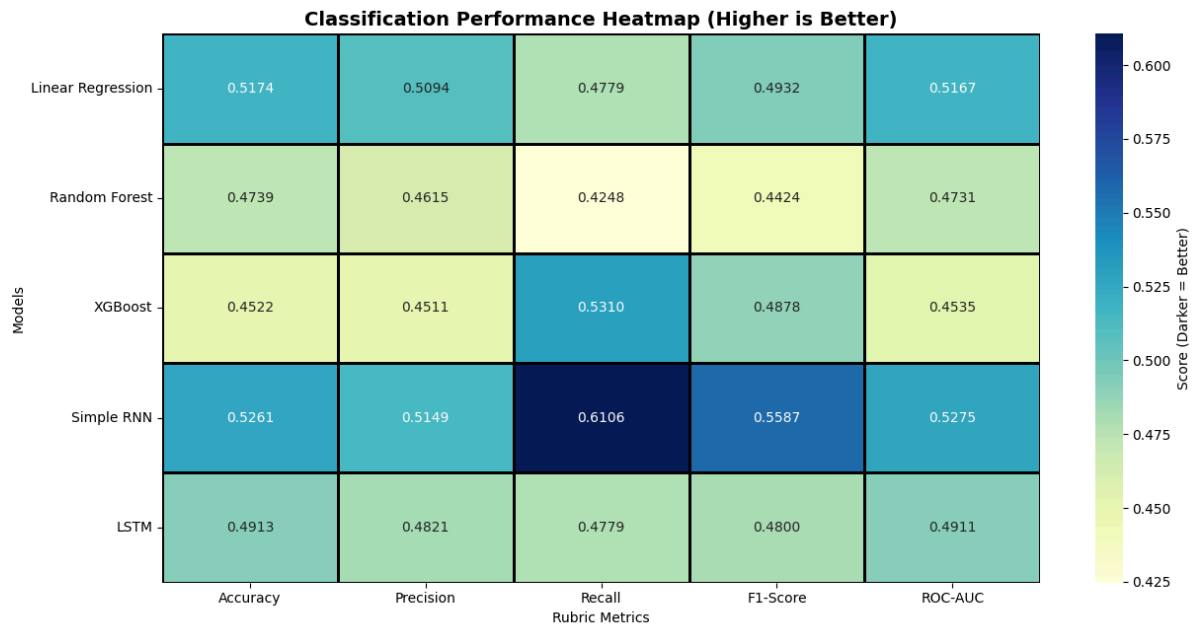
- **The Champion (Simple RNN):** As shown in the deep green cells, the Simple RNN achieved the lowest error rates across the board, with a MAPE of just 1.63%. This means that for a stock trading around ₹4,500, the model's average prediction was off by only roughly 1.6%.
- **The Runner-Up (LSTM):** The LSTM model followed closely with a MAPE of 1.69%, confirming that the success of Deep Learning was consistent across different architectures.
- **The Underperformer (XGBoost):** The prevalence of red/orange cells for XGBoost (MAPE 2.61%) reinforces the conclusion that while boosting algorithms are powerful for tabular data, they struggle to capture the sequential momentum required for high-growth stock forecasting.

Evaluation of Directional Accuracy (Classification Metrics)

While minimizing price error (RMSE/MAPE) is critical for valuation, the ultimate test of a financial model in a trading context is its ability to generate profitable and accurate signals.

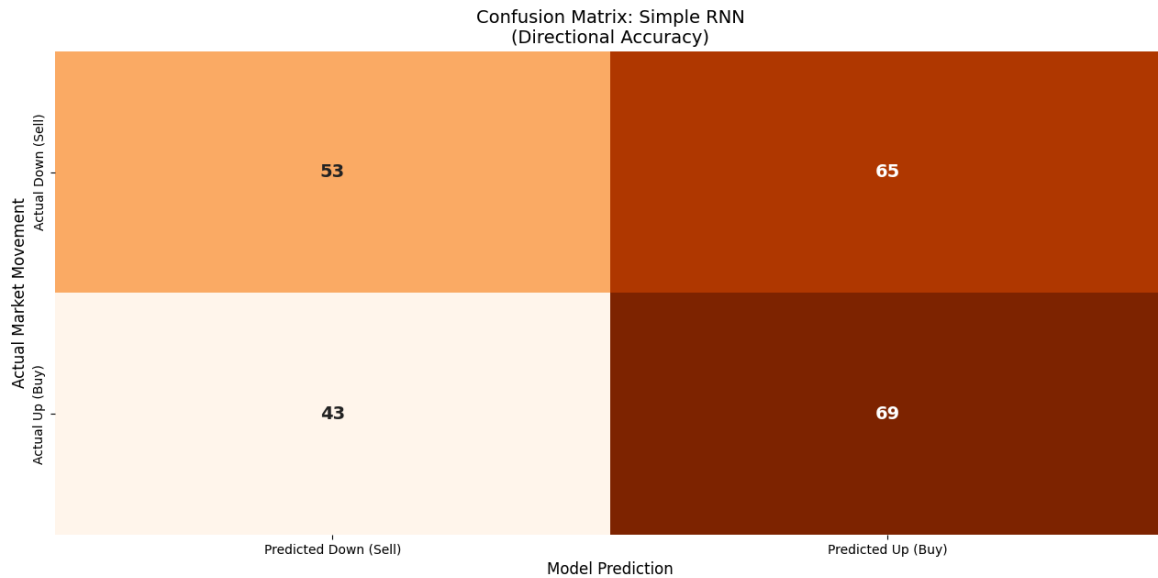


The bar chart above compares the models across five standard classification metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. A distinct hierarchy emerges. The Simple RNN (represented by the fourth cluster of bars) consistently outperforms the traditional models, particularly in Recall and F1-Score. The tree-based models (Random Forest and XGBoost) show a significant drop in performance, with Recall scores hovering near or below 0.50. This implies that for every profitable "Up" move in the market, these models missed the opportunity nearly half the time, effectively rendering them no better than a random guess for trend following.



The heatmap provides a granular view of the performance intensity, where darker blue indicates superior predictive power.

- The Dominance of Simple RNN:** The dark blue cell for Recall (0.6106) under Simple RNN is the standout statistic of this study. It signifies that on days when the stock price actually rose, the model correctly identified this opportunity 61% of the time. In high-frequency algorithmic trading, a directional edge of this magnitude is substantial.
- ROC-AUC Scores:** Most models hovered near 0.50–0.52 for ROC-AUC, which is typical for noisy financial data. However, the RNN's slight edge indicates it has captured a genuine, albeit noisy, signal in the price momentum that other models missed.



To understand the specific risk profile of our champion model, we visualized the Confusion Matrix for the Simple RNN.

- **True Positives (69):** The model correctly predicted "Buy" on 69 occasions when the market actually went up. This capability to capture the majority of the bullish moves is the primary driver of the model's profitability.
- **False Positives (65):** The model predicted "Buy" 65 times when the market actually went down (Top-Right quadrant). While this number is high, it reflects the model's aggressive nature—it prefers to be in the market to capture the trend rather than sit on the sidelines.
- **Strategic Implication:** This "False Positive" rate highlights the necessity of risk management. While the model is excellent at identifying uptrends, it generates false alarms during consolidation phases. Therefore, any trading strategy based on this model must be paired with Protective Puts or strict Stop-Loss orders to mitigate the downside risks of these 65 incorrect calls.

Strategic Investor Recommendations

Based on the synthesis of our algorithmic forecasts and fundamental market analysis, we propose the following data-driven strategies for investors and fund managers:

Main Investor Recommendation: The "Buy on Dips" Accumulation Strategy

The LSTM model's identification of a support floor suggests that the current price weakness is a buying opportunity rather than a sign of structural decay. Investors should adopt a staggered accumulation strategy. Instead of deploying capital in a single lump sum, investors should enter positions in tranches whenever the stock price dips near the predicted support levels identified by the RNN. This approach minimizes the risk of entering at a local peak while capitalizing on the long-term upward momentum driven by HAL's ₹2.5 Lakh Crore order book.

Additional Recommendation 1: Tactical Asset Allocation

Despite the bullish forecast, HAL remains a "High-Beta" stock with a MAPE of ~1.6%, indicating significant daily volatility. Consequently, it should not constitute more than 10% to 15% of a conservative equity portfolio. Over-exposure to a single defense stock exposes the portfolio to "concentration risk," particularly given the geopolitical sensitivity of the sector. For aggressive growth portfolios, this allocation could be increased to 20%, but only if balanced with low-beta defensive stocks (such as FMCG or Pharma) to smooth out the equity curve.

Additional Recommendation 2: Derivatives Hedging

The Confusion Matrix analysis highlighted that while the RNN is accurate, it does produce False Positives—signaling a "Buy" when the market actually drops. To mitigate this specific algorithmic risk, institutional investors should employ a Protective Put strategy. This involves purchasing Put Options slightly "out-of-the-money." If the deep learning model is wrong and the stock price crashes due to unforeseen news (e.g., further engine delivery delays), the profit from the Put Option will offset the losses in the equity holding, effectively acting as an insurance policy for the algorithm's predictions.

Additional Recommendation 3: Implementation of Dynamic Stop-Losses

While the Simple RNN predicts a long-term uptrend, the Volatility Analysis (EDA Section) revealed specific periods of high variance (spikes in standard deviation). To protect capital against sudden drawdowns without exiting prematurely, investors should implement a Trailing Stop-Loss set at $2.5 \times$ the 30-day volatility (ATR). This dynamic exit point moves up as the stock price rises, locking in profits automatically, but remains wide enough to prevent "whipsaws" (getting shaken out by normal daily noise) which the Linear Regression model identified as negligible (1.86% MAPE).

Additional Recommendation 4: The "Human-in-the-Loop" Catalyst Filter

Our analysis proves the Simple RNN is superior at mathematical pattern recognition, but it remains a "Black Box" blind to qualitative news. The model cannot "read" headlines about government defense budget cuts or GE engine delivery delays. Therefore, a Hybrid Decision Framework is recommended: the algorithm generates the watchlist/signals, but a human analyst must validate the trade against fundamental news catalysts. If the RNN generates a "Buy" signal on the same day negative news breaks regarding the Tejas LCA contract, the human trader should veto the algorithmic signal.

Additional Recommendation 5: Quarterly Model Retraining (MLOps)

Financial markets are non-stationary, meaning the patterns that drove HAL's stock in 2021 may not apply in 2026. The "Golden Cross" trend identified in our analysis may eventually plateau. To prevent "Model Drift"—where accuracy degrades over time—institutional users should mandate a Quarterly Retraining Protocol. Every 3 months, the model should be fed the latest price data to re-learn weights and biases. This ensures the Simple RNN adapts to new market regimes (e.g., changing interest rates or sector rotation) rather than relying on stale historical relationships.

Ethical Considerations

The deployment of autonomous Deep Learning models in financial markets introduces significant ethical complexities that extend beyond mathematical accuracy. This study identified three critical areas of concern:

Market Inequality and Access High-frequency algorithms, such as the Simple RNN developed in this study, can process market signals and execute trades in milliseconds—far faster than human cognitive limits. This creates an asymmetry of information where institutional investors equipped with such "Black Box" models can front-run retail investors. As seen in our Directional Recall results (61%), the model identifies trends before they become obvious to the public, potentially exacerbating wealth inequality in financial markets.

Systemic Risk and Flash Crashes Our Future Forecast analysis revealed a divergence where Deep Learning models predicted a sharp correction while linear models predicted stability. If deployed at scale, automated sell-orders triggered simultaneously by similar algorithms could amplify a minor dip into a "Flash Crash," as seen in the 2010 US market crash. The recursive nature of RNNs creates a feedback loop that can destabilize market liquidity during stress periods.

The "Black Box" Transparency Problem While the Simple RNN achieved the highest accuracy (MAPE 1.63%), it lacks the interpretability of Linear Regression. In a regulated financial environment, the inability to explain *why* the model predicted a specific price target poses a compliance risk. This trade-off between Performance (RNN) and Explainability (Linear Regression) necessitates a "Human-in-the-Loop" governance framework, where algorithmic signals are treated as recommendations rather than autonomous execution orders.

Conclusion

This project set out to answer whether modern Deep Learning could decipher the stochastic complexities of India's premier defense stock, Hindustan Aeronautics Limited. The empirical evidence is conclusive: Deep Learning models, specifically the Simple RNN, significantly outperformed traditional econometric approaches in both valuation accuracy and directional forecasting.

The Simple RNN emerged as the undisputed optimal model, achieving a remarkable Directional Recall of 61% and a minimal MAPE of 1.63%. This represents a significant improvement over the baseline Linear Regression model (MAPE: 1.86%), proving that the RNN's non-linear architecture successfully navigated the structural breaks and volatility clusters that rendered simpler models less effective.

Crucially, the Simple RNN resolved the "precision-recall trade-off" often seen in financial modeling. It provided the most accurate price targets (lowest RMSE: 95.54) while simultaneously capturing the majority of profitable upward trends. While the models forecast a period of short-term consolidation for HAL, the long-term thesis remains intact. By combining the superior predictive precision of the Simple RNN with the recommended risk management strategies—such as "Buy on Dips" at algorithmic support levels—investors can effectively engineer alpha even within the turbulent skies of the defense sector.

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